

Tyler Harnish

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EDUCATION

Kean University — Union, NJ

Bachelor of Science in Computer Science, Magna Cum Laude | GPA 3.76 / 4.0

Relevant Coursework: Deep Learning & Applications, Principles of Networking, Computer Security, Database Management Systems, Web Application Development, Object-Oriented Analysis & Design

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, TypeScript, HTML, CSS

Frameworks & Libraries: React, Next.js, PyTorch, torchvision, scikit-learn, Pandas, Tailwind CSS

Databases: MySQL

Tools & Platforms: GitHub, Vercel, Hugging Face Datasets

EXPERIENCE

Technical Assistant

Fall 2024 – May 2026

Nancy Thompson Learning Commons, Kean University

- Supported and maintained a technology-focused lab (V.I.S.E.R), configuring an array of projector displays to meet client needs and providing technical assistance during presentations, university events, and Virtual Reality sessions.

General Laborer

June 2023 – Present

Unique Triangle Apparel, Marlboro, NJ

- Assisted with warehouse operations and inventory organization as well as supporting residents during on-site shopping events at assisted living facilities.

PROJECTS

TrashNet Waste Image Classification Using Deep Learning

Technologies: Python, PyTorch, torchvision, ResNet18, Pandas, Matplotlib, Hugging Face Datasets

- Trained a transfer-learning image classification model on 5,054 TrashNet images across six waste categories: cardboard, glass, metal, paper, plastic, and trash.
- Tuned a pretrained ResNet18 model and evaluated architectural changes, including a deeper classifier head with batch normalization, ReLU activation, and dropout.
- Achieved 95.65% test accuracy and conducted an ablation study to compare the baseline model, architectural modification, and weight-decay regularization.

Portfolio Web Application

Technologies: JavaScript, TypeScript, Next.js, React, PostCSS, Tailwind CSS, Framer Motion, Vercel

- Built a responsive personal portfolio using a component-based architecture to present projects, skills, and education.
- Implemented animated and interactive UI components with Framer Motion and optimized layouts for desktop and mobile devices.
- Deployed the application to Vercel with continuous integration from GitHub to support rapid updates and version control.

Emotion Detection Using Machine Learning (NLP) — Group Project

Technologies: Python, Pandas, scikit-learn, TF-IDF, Logistic Regression, Support Vector Machine

- Preprocessed and analyzed a dataset of 40,000 labeled tweets to perform multiclass emotion classification across 13 sentiment categories.
- Extracted numerical features from text using TF-IDF and trained machine learning models including Logistic Regression and Support Vector Machine.
- Achieved approximately 85% accuracy using SVM on a reduced dataset, demonstrating effective model selection under computational constraints.